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Puerto Princesa City

Power Outage Probability Forecasting with XGBoost and SHAP-Based Feature Analysis Using Historical, Meteorological, Vegetation, and Infrastructure Reliability Data

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Presented by:

Dela Cruz, Reymart

Lagumbay, Eliyahu

Ombion, Kert Ace

Remojo, Rudge Kyle

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TABLE OF CONTENTS

TITLE PAGE.....i

TABLE OF CONTENTS.....ii

LIST OF FIGURES.....iii

LIST OF TABLES.....iii

CHAPTER I: INTRODUCTION

1.1 Background of the Study.....1

1.2 Statement of the Problem.....3

1.3 Research Objectives.....4

1.4 Scope and Limitations….....5

1.5 Significance of the Study.....6

1.6 Operational Definition of Terms.....7

CHAPTER II: REVIEW OF RELATED LITERATURE

2.1 Related Studies.....10

2.2 Related Systems.....12

2.3 Synthesis.....15

CHAPTER III: THEORETICAL BACKGROUND

3.1 Decision Tree-Based Ensemble Learning.....19

3.2 XGBoost (Extreme Gradient Boosting).....21

3.3 Class Imbalance.....26

CHAPTER IV: METHODOLOGY

4.1 Conceptual Framework.....27

4.2 Research Design.....28

4.3 Ethical Consideration.....31

4.4 Dataset Description.....32

4.5 Research Instruments.....34

4.6 Model Development Instruments.....35

4.7 Research Procedures.....36

4.8 Data Analysis Plan.....38

4.9 Work Plan…….....41

REFERENCES.....43

APPENDICES.....47

LIST OF FIGURES

Figure 3.1 Ensemble Learning with Decision Trees Diagram.....20

Figure 3.2 Schematic Illustration of XGBoost.....24

Figure 4.1 Conceptual Framework.....27

Figure 4.2 Gantt Chart.....41

LIST OF TABLES

Table 2.1 Related Studies Comparison Matrix.....11

Table 2.2 Related System Comparison Matrix.....14

Table 2.3 Identified Gaps, and Corresponding Research Objectives……..….17

Table 4.1 List of time-based features.....30

Table 4.2 List of weather features.....31

Table 4.3 Lagged weather and outage-related features.....32

Table 4.4 Vegetation-Related Variable.....32

Table 4.5 Equipment Integrity Proxy Features…….33

Table 4.6 Evaluation Metrics.....37

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Electrical power is vital for economic growth, public security, medical care, communications and daily life. Stable power systems are essential to the operation of residential, commercial, and industrial systems in modern societies. Loss of power services can have a major impact on economic activity and social welfare (International Energy Agency, 2021).

Geographically isolated areas are especially susceptible to outages because of their geographic isolation, limited infrastructure, and generation resources. Puerto Princesa City (PPC) in Palawan has an island based power system that is separate from the Luzon grid. This isolation results in less reliability in the system and greater sensitivity to the variations in demand and supply (Department of Energy Philippines, 2022).

Outages in Puerto Princesa City have a large socio-economic impact. Interruptions impact tourism, business operations and critical services like hospitals and communication. Not only does it make economic growth more difficult, but it also leads to a decrease in residents' quality of life. Various factors that contribute to these include seasonal demand changes, weather, equipment restrictions and vegetation interference and wildlife contact with power lines (Billinton & Allan, 1996).

Environmental, operational, supply and demand factors all contribute to power outages in the Philippines. Local research has revealed that the weather has a significant influence on electricity consumption patterns, particularly temperature (Santos, 2021). Greater temperatures create more consumption, which is also confirmed by the findings of

the Philippine Institute for Development Studies, as cooling appliances are used more during the hot season, thus increasing electricity demands during hot weather, causing power systems with limited backup capacity to be stressed and resulting to frequent power outages (Francisco, 2022).

Outage occurrence and duration is also influenced by weather. Meteorological conditions such as rain, heavy winds, etc. can cause damage to infrastructure and affect transmission systems. Research has demonstrated that rainfall has a significant correlation with outage duration, and temperature is related to higher electricity use (Buencamino, 2024). These findings highlight the importance and feasibility of incorporating weather variables in outage prediction models.

The situation is more serious in the case of island-based systems like Palawan where electricity supply is close to peak demand, particularly during dry seasons. This narrow margin increases system vulnerability, leading to recurring brownouts and service disruptions (Philippine Information Agency, 2026).

In recent years, machine learning has enabled the development of successful methods for power outage prediction. Machine learning models can use the complex relationships between historical outage information, weather information, and time patterns to predict future events (Putra et al, 2025). Such models can be used for nonlinear relationships and large data sets, which are good for real-world energy applications.

Tree-based ensemble models like Random Forest, LightGBM and XGBoost have been found to have strong predictive performance for classification and forecasting, especially in capturing complex interactions and non-linearity (Chen & Guestrin, 2016). However, their application in small, isolated grid systems remains limited.

Recent studies focus on large-scale grids and disaster-related outages; these conditions do not fully represent the setup of Puerto Princesa City. Puerto Princesa City operates under an island-based power system where outages does not only occur during extreme weather events but also under routine or moderate conditions that may be influence by supply-demand limitations, local weather, equipment constraint and environmental factors. Therefore, this study aims to develop a XGBoost-based predictive model using historical outage logs, weather data, and temporal features to forecast short-term power outages and support proactive decision-making for local utility providers.

weather

1.2 Statement of the Problem

Power outages are a recurring concern in Puerto Princesa City, and the local utility lacks a practical, data-driven way to anticipate them. Most existing research focuses on large, data-rich grids or extreme disaster scenarios, which does not reflect the city's island-based distribution context. This shows that there is a need for a machine learning approach that can effectively utilize the data available within the distribution context. Specifically, this study seeks to answer the following questions:

1. How can historical outage logs, and features such as meteorological variables, temporal patterns, area vegetation, and infrastructure reliability be integrated to develop an XGBoost-based probability forecasting for Puerto Princesa City?

2. How well does the developed prediction model perform in forecasting power outages, as evaluated by standard performance metrics such as AUC-ROC, F1-score, precision, recall, calibration analysis, and Brier Score ?

3. Which meteorological, temporal, infrastructure, and vegetation-related features are identified by SHAP (SHapley Additive exPlanations) as the most significant to

predicted likelihood of the power outage in the selected reclosers in Puerto Princesa City?

4. To what extent do lagged features and cumulative outage frequency contribute to the accuracy of predictive models?

1.3 Research Objectives

This study aims to develop an effective predictive approach in forecasting power outages in Puerto Princesa City, specifically by leveraging historical outage, vegetation and weather-related data. Specifically, it focuses on utilizing the relevant features to train and validate a machine learning model, and evaluate its performance. The specific objectives of this study are as follows:

1. To develop and optimize an XGBoost-based probability forecasting model by systematically integrating standardized historical outage records, meteorological variables, engineered temporal and infrastructure reliability features, and vegetation-related variables to capture localized nonlinear relationships influencing power outages in Puerto Princesa City.

2. Evaluate the predictive performance of the developed model using standard machine learning metrics, specifically AUC-ROC, F1-score, precision, and recall, calibration analysis, and Brier Score to determine its effectiveness in forecasting power disruption.

3. Identify and interpret the key features that contribute most to predicted outage likelihood to the selected areas by applying SHAP (SHapley Additive exPlanations) analysis to the trained model.

4. Assess the impact of engineered features, such as lagged meteorological data and rolling outage frequency, to determine their contribution to the model's overall accuracy.

1.4 Scope and Limitations

The aim of the proposed study is to develop and test a predictive model to estimate power outage probability in Puerto Princesa City based on historical outage data, weather data, temporal features, structural reliability features, and vegetation-related variables using machine learning techniques.

The study applies the predictive performance of XGBoost, a gradient boosting ensemble algorithm, in forecasting power outages and is evaluated using standard machine learning evaluation metrics such as AUC-ROC, F1-score, precision, and recall. In addition, the probability calibration is assessed through reliability diagram analysis and Brier Score, while SHAP (SHapley Additive exPlanations) analysis is used to interpret and identify the features that contribute most to outage probability predictions.

The dataset utilized for the proposed study consisted of historical outage records, containing time-based variables such as date and timestamps, panning from September 2022 to 2026 from the local distribution company. Vegetation-related data includes the NDVI values and vegetation landcover surrounding each coverage area how it changes over time. Weather-related factors include, but not limited to, temperature, rainfall, wind speed, and other available meteorological data. Thus, the study accounts for both the operational and environmental factors in an attempt to capture patterns that can be expected to affect outage occurrences, both in normal operation and during weather-related disturbances.

Limitations of the Study

This study is subject to several limitations. First, the accuracy of the prediction model depends on the quality, completeness, and availability of the historical outage, weather datasets, inconsistent reporting, or limited historical records may affect the reliability and performance of the model.

Second, the study is limited to data collected within a specific geographic location and distribution system. The study only covers Puerto Princesa City, using outage records from the Palawan Electric Cooperative (PALECO) for September 2022 to 2026. The city is not treated as a single spatial unit, rather the model estimates the possible outage probability and which specific areas will be affected. As a result, the findings and model performance may not be directly generalizable to other regions with different infrastructure conditions, climate patterns, or operational practices.

Additionally, while the study includes weather-related variables, it does not focus exclusively on extreme weather events such as typhoons or large-scale natural disasters. Instead, the model is designed to capture both disruptions occurring under normal to moderate conditions and severe weather-influenced disruptions within the available datasets. Therefore, the model's performance during rare, large-scale catastrophic events may still be limited.

Finally, this study is explicitly model-centric in scope. The primary output is a trained and evaluated XGBoost prediction model, together with the results of the associated performance metrics and SHAP-based feature analysis. The study does not cover the development of a deployable software application or operational forecasting system, which could be a practical extension of the study, but is beyond the scope of this research.

1.5 Significance of the Study

This study holds significant value in addressing the persistent challenges faced by many communities due to the unpredictability of power interruptions. By analyzing historical outage and weather data, this research aims to contribute a data-driven approach in predicting possible power disruptions before they occur, providing knowledge and practical solutions. In addition, the results of this study would be beneficial to the following:

Power Utilities. The predictive model developed in this proposed study can equip these organizations with actionable information, enabling them to make better and informed decisions, allocate resources more efficiently, and reduce response times during outage events. Rather than relying solely on reactive strategies, utility providers can transition toward a more proactive operational framework, with pattern recognition and weather-correlated risk assessment. The result may help utility providers understand outage patterns within the selected reclosers.

General Public. Frequent and unannounced power outages disrupt daily life, affect productivity, compromise the operation, and impose economic burdens on households and small businesses. A reliable forecasting system, informed by the findings of this study, can enable utility companies to issue advance advisories, giving communities time to prepare.

Future Researchers. This research adds a context-sensitive contribution to the growing body of knowledge on applied machine learning in energy systems and provides a replicable methodological framework that future researchers may adapt for other cities or regions facing similar challenges.

1.6 Operational Definition of Terms

For better clarification and understanding of the information conveyed in the study, the following terms are defined.

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) - a threshold independent metric that quantifies a model's ability to discriminate between positive and negative classes across all classification thresholds.

Binary Classification - a supervised machine learning task in which a model assigns each observation to one of two mutually exclusive categories.

Brier Score - a metric that measures the mean squared error between predicted probabilities and actual binary outcomes, with values ranging from 0 (perfect) to 1 (worst).

Distribution Circuit - a segment of the electrical distribution network that delivers power from a substation to end consumers within a defined coverage area. This study is delimited to a single distribution circuit operating within Puerto Princesa City, Palawan, which covers multiple barangays.

Gradient Boosting - an ensemble machine learning technique that constructs a strong predictive model by sequentially combining multiple weak learners. It forms the algorithmic foundation of XGBoost.

Historical Outage Logs - recorded data documenting past instances of power interruptions, including details such as the date, time, duration, location, and cause of each outage.

Infrastructural Reliability - the capability of physical systems, such as electrical distribution networks, to consistently perform their required functions without failure under specified conditions over a given period. It focuses on maintaining continuous uptime, operational stability, and the consistent delivery of services.

Machine Learning - a subfield of artificial intelligence concerned with algorithms that learn patterns from data to make predictions or decisions without being explicitly programmed for

each outcome. In this study, machine learning is applied to historical outage and meteorological data to forecast the probability of power outage occurrence.

Meteorological Data - also referred to as weather data is quantitative measurements of atmospheric or weather conditions collected over a specific period. In this study, meteorological data refers to daily weather records from OpenMateo, including temperature, rainfall, precipitation, wind speed, wind gusts, humidity, and weather code.

Power Outage - an unplanned interruption in the supply of electricity to a specific area or set of consumers, typically caused by equipment failure, severe weather events, or other unforeseen circumstances.

Predictive Model - Refers to the mathematical or algorithmic framework trained on existing data to forecast future outcomes.

Probability Calibration - the degree to which a model's predicted probability outputs correspond to the true observed frequencies of the outcome.

SHAP (SHapley Additive exPlanations) - a game-theoretic framework that attributes each model prediction to the contribution of individual input features.

Temporal Lag Features - variables constructed by referencing the values of a prior time period as predictors for a current observation. In this study, lag features derived from historical outage records encode recent outage activity as input signals for outage forecasting.

XGBoost (Extreme Gradient Boosting) - a regularized, scalable implementation of gradient boosting with improved predictive accuracy and overfitting control. It is the predictive model developed and evaluated in this study.

CHAPTER II

REVIEW OF RELATED LITERATURES AND SYSTEMS

This chapter examines existing research on power outage forecasting to understand the methods, results, and limitations of previous work. By evaluating these studies, the researchers identify the technical and theoretical gaps that this project addresses.

2.1 Related Studies

Studies on power outages used different types of data and many show that weather conditions and historical records are commonly used to analyze outage behavior. These variables are not treated as direct causes on their own but are used as input in models to identify patterns and improve prediction performance.

Several studies have used weather- or climate-related variables in developing predictive models. Talukdar & Toweh (2024) included meteorological variables in their machine learning models, finding that while weather is relevant to outage prediction, it may not be sufficient on its own if the grid infrastructure is already resilient.

Historical records like start times, duration, and frequency help identify patterns in power interruptions. Saidi et al. (2021) used lightning data and tripping reports to predict outages using Logistics Regression and Decision Tree models. Their results showed that the Fine Tree model was the most suitable model based on AUC-ROC comparison. This highlights the importance of using outage records and weather factors in predicting power outage occurrence.

Other studies combine weather and historical outage data. Putra et al. (2025) built an early warning system using Random Forest, XGBoost, AdaBoost and LightGBM. They

found that LightGBM with SMOTE achieved a recall of 0.99. Wang et al. (2022) also used weather and outage records and found that this combination improved prediction accuracy.

Avcı (2024) tested a hybrid model using outage records, weather data, and geographical information. This system combined neural networks and tree-based methods to boost prediction accuracy. However, this method relies on detailed Outage Management System (OMS) data, which provides high-resolution insights into grid equipment and real-time network topology. Such detailed datasets are not available for isolated power systems like the one in Puerto Princesa City. Because of this, the researchers cannot apply Avcı's complex model to the Puerto Princesa City grid without the infrastructure data the original study requires.

Table 2.1 Related Studies Comparison Matrix

Author	Year	Method	Findings	Relation to Study
Talukdar & Toweh	2024	ANN model using climate factors such as temperature, precipitation, humidity, wind speed, and solar radiation	ANN models were not effective in predicting outages using weather factors alone. The authors said that this may be due to skewed datasets and the selected locations being more resilient to weather-related outages	Supports the idea that weather data is relevant, but weather variables alone may not be enough. This supports combining weather data with historical outage data.
Saidi et al.	2021	Logistic Regression and Decision Tree using lightning data, and tripping reports in Malaysia	Fine Tree was the most suitable model for power outage prediction based on AUC/ROC comparison	Uses weather-related factors and utility tripping/outage reports in Malaysia
Putra et al.	2025	Random Forest, XGBoost, AdaBoost, and LightGBM using historical weather and outage data in Indonesia	LightGBM with SMOTE achieved the highest recall of 0.99	Supports combining historical weather and outage data in Indonesia.
Wang et al.	2022	Regression, Random Forest, XGBoost, and time-series models using historical outage data, weather attributes, and ERCOT	Models using both weather features and ERCOT data produced lower RMSE and higher prediction accuracy than models using only one	Shows that combining outage data and weather features improves prediction

		energy market data	feature set	
Avcı	2024	Hybrid AI model using OMS records, weather forecasts, and geographical data	Hybrid models improve prediction accuracy and highlight the role of wind speed and storms in outages	Support uses of combining historical outage records and weather data.

Table 2.1 shows the reviewed literatures examined and highlights the features of the different studies that implement diverse computational frameworks, including supervised machine learning, ensemble methods, and hybrid deep learning architectures, which integrates weather variables with historical data of outages to predict power disruptions as a baseline for this study.

2.2 Related Systems

This section evaluates relevant systems that have pioneered specific methodologies in outage forecasting. It showcases the development of computational frameworks for predicting power grid disruptions, from simple statistical regressions to complex deep learning architectures.

According to the study by Liaqat et al. (2025), a Hierarchical RNN-LSTM framework was developed to predict multi-class outages in microgrids, focusing on capturing temporal dependencies at various scales. This system evaluates multi-scale temporal patterns by structuring stacked LSTM layers to operate at 5-minute, hourly, and daily resolutions, allowing the model to distinguish between diverse disruption causes such as equipment failures, cyber-attacks, and weather-induced events.

The proposed RNN-LSTM model utilizes high-frequency telemetry data, including solar irradiation, load demand, and power metrics, and integrates a time-aware embedding strategy to handle categorical features. By leveraging gated memory units to retain critical

patterns over extended sequences, the system achieved a high classification accuracy on both national datasets and real-time microgrid telemetry.

According to the study by Sharma et al. (2023), a methodology was presented to forecast weather-related distribution outages using weighted logistic regression to address extreme data imbalance. The system focuses on one-day-ahead hourly forecasts by determining weights for outage and non-outage hours based on the reciprocals of their frequencies in historical training sets.

The proposed weighted model uses actual meteorological observations, such as wind speed gusts, precipitation, and snowfall, to produce binary classification outputs at both city-wide and distribution substation levels. Results indicated that the inclusion of reciprocal weighting significantly outperformed ordinary logistic regression by preventing the model from under-fitting the minority outage class.

According to the study by Wang et al. (2024), deep learning-based conditional and unconditional Multi-Layer Perceptron (MLP) structures were developed to predict hourly outage probabilities at the census tract level. The system treats weather data as a base input while using location characteristics, such as power infrastructure maps and structure ages, as a modulating condition to reflect varying local resilience levels.

The proposed conditional MLP utilizes a two-branch architecture to process weather features and community demographics independently before fusing them to predict normalized customer outages. The research found that incorporating socio-economic indicators effectively reduced error rates, identifying factors like household income and infrastructure density as significant predictors of outage frequency.

According to the study by Lee et al. (2025), nonparametric machine learning models, including XGBoost, LightGBM, and CatBoost were utilized to predict power outages during extreme winter storms in the southern United States. The system identifies population and atmospheric pressure as key predictors, reflecting the supply-demand imbalances that trigger large-scale disruptions during extreme cold waves.

The proposed XGBoost baseline model achieved high predictive accuracy by utilizing Bayesian optimization for hyperparameter tuning and data augmentation via Gaussian noise injection. Partial dependence analysis revealed that dew point and population size have the most significant impact on model forecasts, particularly in regions unaccustomed to severe winter conditions.

According to Avcı (2024), a hybrid artificial intelligence approach was proposed to improve electricity outage prediction. This system utilizes historical outage records, meteorological data, and geographical information, to identify patterns that influence electricity disruptions. The system utilizes a combination of neural networks (CNN and RNN)) and tree-based models (XGBoost), which outperformed individual models.

Table 2.2 Related System Comparison Matrix

Features	Liaqat et al. (2025)	Sharma et al. (2023)	Wang et al. (2024)	Lee et al. (2025)	Avcı (2024)	Proposed System
Historical Outage Data	✓	✓	✓	✓	✓	✓
Meteorological Data	✓	✓	✓	✓	✓	✓
Handles Temporal Patterns	✓	✓	✓	×	✓	✓
Covers Routine	✓	×	×	×	✓	✓

Outages						
Covers Weather Outages	✓	✓	✓	✓	✓	✓
Focus only on Extreme Weather Context	×	✓	×	✓	×	×

Table 2.2 shows the reviewed studies examined and analyzes the features of diverse computational frameworks presented, including sequence-aware deep learning architectures, hybrid ensemble methods, and probabilistic regression models, utilized to predict power grid disruptions across various contexts, as baseline for the proposed system.

2.3 Synthesis

Across these reviewed studies, there is consistent observation that combining meteorological and historical outage information improves model performance. Weather variables provide information about environmental conditions, while historical outage data reflects patterns from past events. When both are used together, these variables can enhance the ability of machine learning models to capture relationships between variables, thereby enhancing accuracy. In addition, studies and systems that integrate multiple data sources, including infrastructure, socio-economic, and geographical information, perform better than single-input models.

Across the reviewed systems, a range of computational approaches has been applied to predict power outages, from statistical models to advanced deep learning architectures. Models such as the hierarchical RNN-LSTM (Liaqat et al., 2025) demonstrate strong capability in capturing temporal dependencies across multiple time scales, making them effective for analyzing complex and sequential outage patterns. Similarly, deep learning models such as the conditional MLP (Wang et al., 2024) enhance prediction accuracy by

incorporating multiple data sources, while ensemble and nonparametric models such as XGBoost, LightGBM, and CatBoost (Lee et al., 2025) provide flexibility in handling nonlinear relationships. Simpler approaches such as weighted logistic regression (Sharma et al., 2023) address specific challenges like data imbalance but are limited in modeling complex interactions. Meanwhile, hybrid approaches (Avcı, 2024), which combine neural networks and tree-based models, further improve performance by leveraging the strengths of multiple algorithms.

However, many of the reviewed studies focus on large power systems or extreme weather events such as storms and disasters. There is limited research that focuses on isolated power systems such as Puerto Princesa City. Some advanced studies depend on real-time sensors, infrastructure maps, OMS records, geographic data and socio-economic data which are not always available in small power systems.

Because of these limitations, the proposed study focuses on frequent and recurring power interruptions outage in selected distribution circuit in Puerto Princesa City ,rather than focusing only on large-scale outage events but the proposed study is not only limited to normal operating conditions, since it may also cover outage data that happens on moderate to severe conditions. This study uses practical and commonly available data, including weather variables and historical outage data. Historical outage data are recorded by power utilities, while meteorological data can be obtained from public APIs.

Overall, the literature shows that meteorological variables along with historical outage are widely used for outage prediction. While additional data may improve prediction, using available and practical data is still a valid and accepted approach. This makes the selected variables, and the use of an ensemble tree-based model, suitable for developing a practical

and implementable outage prediction system, especially in systems where real-time monitoring data is limited.

Area	**Identified Research Gap**	**Implication to the Study**	**Identified Problem in Project Context**
Weather-Related and Utility Tripping Reports. Saidi et al. (2021)	Uses weather factors and utility records but it does not focus on localized distribution systems.	The proposed study can use utility outage records and weather data as practical input sources for local outage forecasting.	Obj. 1 & 2: Use outage records and weather data to develop and evaluate a predictive model in a selected distribution system in Puerto Princesa City.
Weather-Based Outage Prediction. Talukdar & Toweh (2024)	Weather variables may not be enough to predict outages in areas with limited history data and resilient infrastructure.	Need for expanding the data sources and long time-frame of historical data and equipment age if available.	Obj. 1**:** Develop an XGBoost model using historical outage logs, meteorological variables, and temporal features in a selected distribution system in Puerto Princesa City.
Historical Weather and Outage Data for Early Warning. Putra et al. (2025)	Focuses on extreme weather events, which may ignore the possible cumulative stress and antecedent effects caused by weather variables	Opportunity to adopt the idea of combining historical weather and outage data, but apply it to a localized setting.	Obj. 1 & 2: Develop and evaluate a machine learning model using historical outage and meteorological data in a selected distribution system in Puerto Princesa City.
Historical Outage + Weather + Time-Series Methods. Wang et al. (2022)	Conducted in major metropolitan areas and used additional markets that do not reflect the local operating conditions of Puerto Princesa City.	Opportunity to focus on practical data sources available locally such as outage logs and weather data.	Obj. 1 & 2: Build and evaluate a practical outage forecasting model using available historical and meteorological data in a selected distribution system in Puerto Princesa City.
Hybrid AI Using OMS, Weather, and Geographic Data. Avcı (2024)	The system depends on detailed OMS records, geographic data, and infrastructure-level information, which is hardly accessible in local settings.	The proposed study addresses this practical data limitation by using commonly available outage logs and public meteorological data.	Obj. 1: Develop a practical XGBoost model using accessible data sources in a selected distribution system in Puerto Princesa City.
Multi-Scale Temporal Outage Prediction**.** Liaqat et al. (2025)	The system requires high-frequency telemetry and microgrid operational data which is not available in small distribution systems..	The proposed study should still capture temporal patterns, but through simpler engineered temporal	Obj. 1 & 3: Use temporal features and interpret their contribution to outage likelihood.

		features such as hour, month, season, and lagged variables.	
Weather-Related Distribution Outage Forecasting. Sharma et al. (2023)	Study used a linear classification model, which may be limited in capturing nonlinear relationships among weather, temporal, and historical outage features	Opportunity to use a model capable of capturing non-linear relationships.	Obj. 1**:** Develop an XGBoost model using historical outage logs, meteorological variables, and temporal features in a selected distribution system in Puerto Princesa City.
Hourly Outage Probability with Weather, Infrastructure and Socio-Economic Data. Wang et al. (2024)	The study depends on infrastructure and socio-economic data but infrastructure related information is not available in public.	Need for implementable approach and the use of practical data available.	Obj. 1 &2: Develop outage probability forecasting using available data and evaluate the reliability of probability outputs.
Nonparametric ML for Winter Storm Outages. Lee et al. (2025)	The study is focused on extreme winter storms and a non-tropical context which differs from the applicability of variables in tropical localized settings.	The opportunity to apply XGBoost for outage prediction but apply it to a tropical setting.	Obj. 1, 2 & 3: Use XGBoost for forecasting, evaluate model performance, and interpret key contributing features.

Table 2.3 Identified Gaps, and Corresponding Research Objectives

Table 2.3 summarizes the reviewed studies by showing their gaps, implications and connections to the objectives of the proposed study. The table highlights that reviewed studies support the use of historical outage data and meteorological variables and machine learning for outage prediction. Many rely on large distribution systems, extreme weather, and well documented grid history settings. This supports the need for a localized outage forecasting model using practical data available for selected distribution circuits in Puerto Princesa City.

CHAPTER III

THEORETICAL BACKGROUND

This chapter presents the theoretical framework and concepts relevant to the study. It discusses the machine learning theories, ensemble learning techniques, mathematical foundations, and predictive modeling approaches used in developing a power outage forecasting model based on historical outage records, meteorological variables, and temporal features.

3.1 Decision Tree-Based Ensemble Learning

Decision tree-based ensemble learning is a robust predictive modeling paradigm that aggregates the outputs of multiple individual decision trees to enhance predictive accuracy and model stability. At its core, a decision tree serves as a non-parametric supervised learning method that partitions the feature space into local regions through a series of recursive binary splits based on feature thresholds (Breiman et al., 1984). In ensemble learning, these base learners are combined through two primary strategies: Bagging and Boosting. Bagging, exemplified by the Random Forest algorithm, constructs multiple independent trees using bootstrap samples and averages their predictions to mitigate variance and prevent overfitting. On the other hand, boosting builds trees sequentially, where each new tree attempts to correct the errors made by previous ones.

Conversely, Boosting algorithms, such as XGBoost or LightGBM, develop trees sequentially, where each successive iteration focuses on minimizing the residual errors of the preceding ensemble (Chen & Guestrin, 2016). This collective approach allows the model to capture complex, high-dimensional patterns that a single learner might overlook.

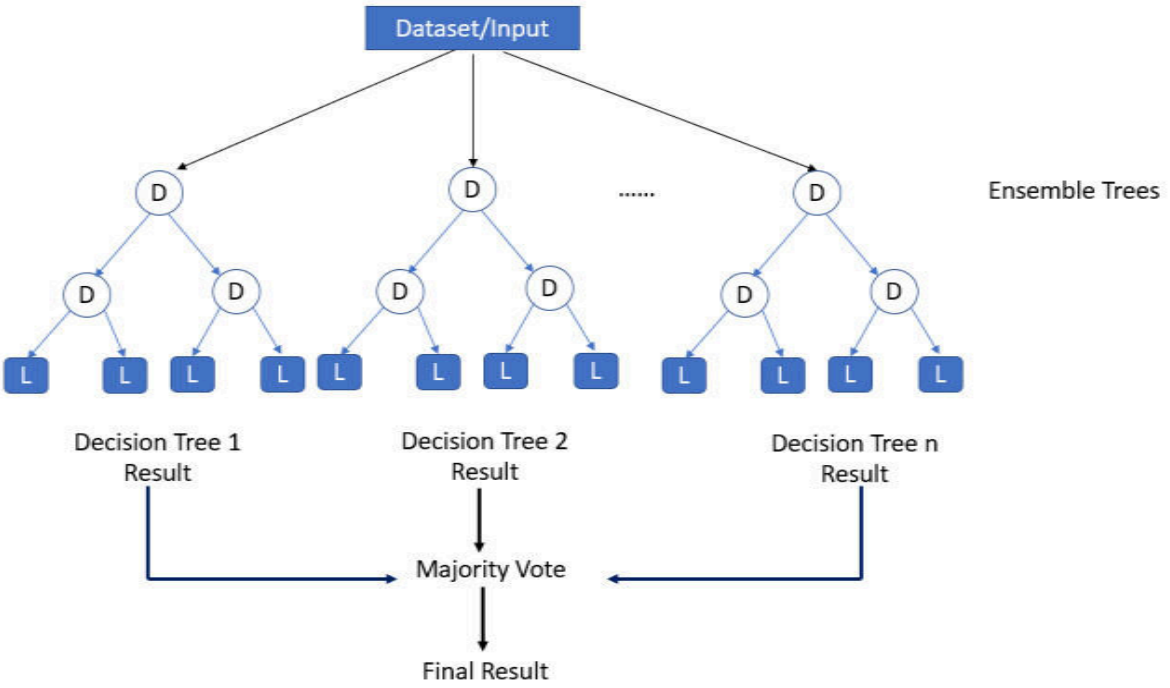


Figure 3.1 Ensemble Learning with Decision Trees Diagram

Figure 3.1 shows the architectural workflow of a decision tree-based ensemble learning model, where a primary dataset is processed through multiple parallel decision trees. Each individual tree progresses from decision nodes (D) to leaf nodes (L), generating independent results that are then funneled to a majority vote. This aggregation mechanism synthesizes the various tree outputs into a single, cohesive result, illustrating how the model improves predictive accuracy and stability by relying on a collective consensus.

The general prediction function for a tree-based ensemble of ***M*** trees for an input vector ***x*** is typically expressed as:

$$\hat{y}(x) = \sum_{m=1}^{M} w_m T_m(x)$$

Where:

- **$\hat{y}(x)$** = The final predicted output.
- ***M*** = The total number of trees in the ensemble.

- $\boldsymbol{T_m(x)}$ = The individual prediction from the $\boldsymbol{m}$-th decision tree.
- $\boldsymbol{w_m}$ = The weight assigned to the $\boldsymbol{m}$-th tree (often $\boldsymbol{1/M}$ for simple averaging).

In the context of this study, decision tree-based ensemble learning is suitable because power outage occurrence is influenced by multiple interacting variables, including weather conditions, historical outage behavior, and temporal patterns. Ensemble learning techniques can capture nonlinear relationships and complex feature interactions, making them effective for short-term outage forecasting in island-based power systems such as Puerto Princesa City.

3.2 XGBoost (Extreme Gradient Boosting)

XGBoost (Extreme Gradient Boosting), is an ensemble machine learning algorithm based on decision trees. It works like the traditional Gradient Boosting algorithm, where multiple weak learners (decision trees) are created sequentially and combined together to produce a strong predictive model. Each new decision tree is trained to correct the errors made by the previous trees, allowing the final model to produce accurate predictions.

XGBoost was introduced by Chen and Guestrin (2016) as a scalable tree boosting system. It is used in classification and prediction tasks because it can handle structured data, capture complex nonlinear relationships, and reduce overfitting through regularization. These characteristics make it suitable for analyzing complex relationships between historical outage records, meteorological variables, and temporal features.

The prediction model of XGBoost is based on an additive structure, where the final prediction is formed by summing the outputs of several decision trees. Each tree provides a partial contribution to the final model output.

$$\hat{y}i = \sum_{k=1}^{K} fk(xi)$$

Where:

- $\hat{y}i$ = predicted output for the ith data record
- xi = input feature vector of the ith data record
- K = total number of decision trees in the model
- $fk(xi)$ = prediction of the kth decision tree for the ith data record

This equation shows that XGBoost does not rely on a single decision tree. Instead, it combines the outputs of multiple decision trees to produce a stronger prediction. For probability-based prediction, the raw model output can be transformed into a probability value using the logistic function:

$$pi = \frac{1}{1 + e^{-F(xi)}}$$

Where:

- pi = predicted probability score
- $F(xi)$ = raw prediction score produced by the model
- e = Euler's number

The probability value ranges from 0 to 1. To make the result easier to interpret, the probability can be converted into a percentage:

$$Prediction\ Percentage = pi \times 100$$

For example, if the model produces a probability value of:

$$pi = 0.85$$

Then:

$$0.85 \times 100 = 85\%$$

This means that the model estimates a 85% likelihood of the event occurring based on the given input features.

Objective Function of XGBoost

XGBoost trains the model by minimizing an objective function. The objective function contains two main parts: the loss function and the regularization term.

$$L(\phi) = \sum_{i=1}^{n} l(yi, \hat{y}i) + \sum_{k=1}^{K} \Omega(fk)$$

Where:

- L(ϕ) = overall objective function
- $l(yi, \hat{y}i)$ = loss function that measures prediction error
- yi = actual output
- $\hat{y}i$ = predicted output
- $\Omega(fk)$ = regularization term
- fk = the kth decision tree
- n = total number of data records
- K = total number of decision trees

The first part is the loss function which measures the difference between the actual and predicted output. The second part is the regularization term, which helps penalize the complexity of the decision trees to prevent overfitting. Overfitting happens when the model performs well on the training data but performs poorly on new or unseen data. This helps improve the model's generalization performance on unseen data.

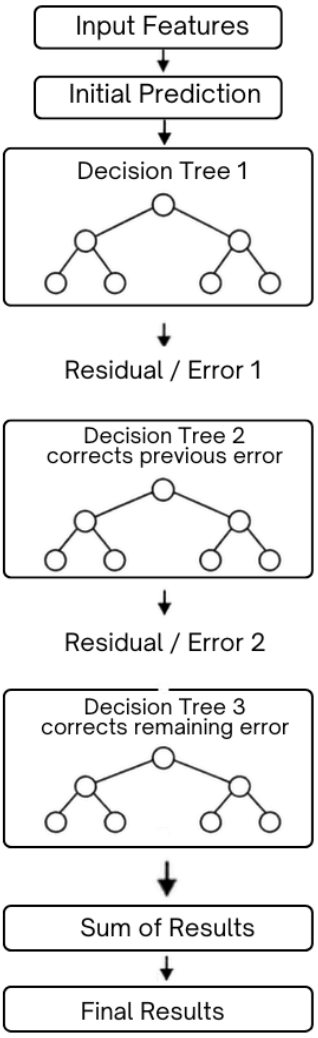


Figure 3.2 Schematic illustration of XGBoost.

Figure 3.2 illustrates the workflow of the XGBoost model for outage likelihood prediction. Input features include meteorological variables, temporal variables and historical outage information. XGBoost first generates an initial prediction and then sequentially builds decision trees, where each tree minimizes the prediction errors of the previous trees. The model sequentially builds decision trees, where each tree minimizes the prediction errors of the previous trees. The sum of outputs are transformed using the logistic function to generate an outage probability score that represents the likelihood of power interruption occurrence.

The probability output of XGBoost supports the objective of helping utility providers make data-driven decisions. Instead of giving only an "outage" or "no outage" result, the model can provide an estimated outage likelihood percentage. This allows the result to be interpreted as a risk level, which can be useful for planning and early response.

SHAP Explainability

SHAP (SHapley Additive exPlanations) is a model interpretation method that assigns an importance value to each feature used for model prediction. It shows how each input feature contributes to the predicted output. SHAP was introduced as a unified framework for interpreting model predictions by Lundberg and Lee (2017).

In this study, SHAP will be used to interpret the output of XGBoost model by identifying how each input feature contributes to the predicted outage likelihood. SHAP will help determine which weather, outage-related, temporal feature and vegetation-related variables have the most contribution to the model's predictions.

The SHAP model can be expressed as:

$$f(x) = \phi 0 + \sum_{i=1}^{M} = \phi i$$

Where:

- $f(x)$ = model prediction
- $\phi 0$ = base value
- ϕi = is the SHAP value of feature i
- M = total number of features

3.3 Class Imbalance

Class Imbalance Problem refers to a challenging issue in machine learning where there is a disproportionate ratio of instances between different classes, leading to biased models that misclassify the minority class and result in poor classification performance (Khan et al., 2024). In the context of the study, outage-free days typically outnumber outage days, as even disruption-prone distribution circuits operate normally for portions of the observation period.

This imbalance poses a fundamental theoretical problem for standard machine learning training procedures. Under imbalanced conditions, a naive classifier that predicts the majority class for all observations, predicting no outage every day, can achieve high raw accuracy while providing no predictive value whatsoever. Standard accuracy is therefore a theoretically inappropriate primary metric for this study and is not reported as a measure of model quality.

In this study, XGBoost addresses class imbalance through the scale_pos_weight hyperparameter, which assigns a multiplicative weight to gradient contributions from positive class observations. A typical value is:

$$scale_pos_weight = \frac{count(y = 0)}{count(y=1)}$$

This effectively increases the penalization of false negatives or days on which an outage occurs but the model predicts none relative to false positives. In utility grid management, failure to anticipate an outage (false negative) is generally more consequential than a false alarm (false positive), providing additional applied justification for this weighting strategy.

CHAPTER IV

METHODOLOGY

This chapter presents the methodology used in the study, including the procedures for data collection, system development, and evaluation. It also explains the approaches used to ensure the reliability, validity, and overall effectiveness of the research.

4.1 Conceptual Framework

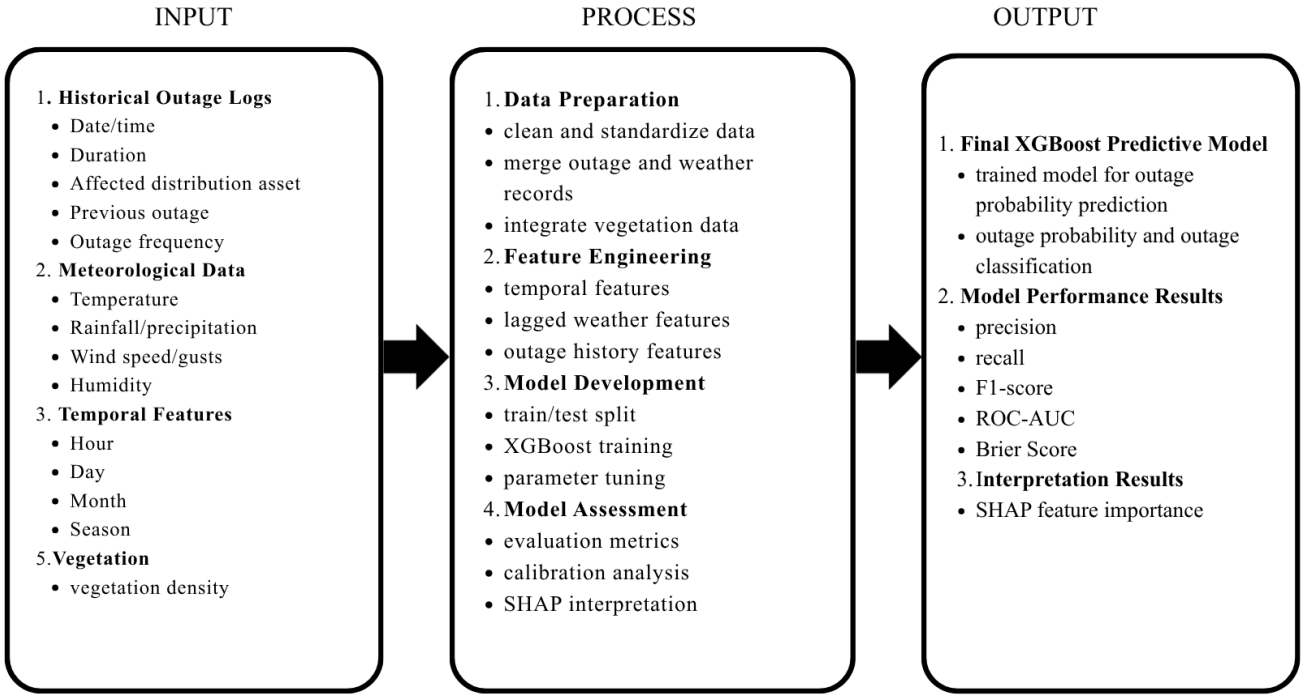


Figure 4.1 Conceptual Framework

The conceptual framework of the study follows the Input-Process-Output model. The input consists of historical outage logs, meteorological data, temporal features, and vegetation related variables. The historical outage logs from PALECO will be organized based on WESCOM Circuit distribution categories, including primary poles, secondary poles, primary lines, secondary lines, and transformers. These inputs serve as the main variables for forecasting power outages. It will then be prepared and processed through data preparation, feature engineering, model development, and model assessment. During this stage, the data

will be cleaned, merged, transformed into useful features, and used to train and evaluate the XGBoost model.

The output of the study is the final XGBoost predictive model. It provides an outage likelihood percentage based on the given input conditions. The study will also produce model performance results using precision, recall, F1-score, ROC-AUC, and Brier Score. SHAP interpretation will also be used to identify the most important features that contribute to the model's outage predictions.

4.2 Research Design

This study adopts a quantitative and applied research design, grounded on the development of a predictive model. According to Creswell and Creswell (2018), quantitative research is an approach for testing objective theories by examining the relationship among variables, which can be measured and analyzed using statistical procedures. This design is suitable for this study as it uses systematic collection, processing, and analysis of numerical data to achieve measurable and statistically evaluable results, such as historical power outage logs and meteorological measurements. The applied nature of the study is evident in the practical focus of the research; this study not only intended to contribute to the theoretical knowledge base, but also to create a useful predictive tool that can be applied to a real-world operational problem that is currently being experienced by local distribution utilities.

The design also aligns with previous research in the field of power outage prediction and machine learning, which also used a quantitative and experimental approach to model the relationship between meteorological variables and power grid disruptions by using measurable performance indicators (Talukdar & Toweh, 2024). Furthermore, this study develops and evaluates a machine learning model using the Extreme Gradient Boosting

(XGBoost) algorithm, which is trained using historical outage, meteorological data and vegetation-related variables to predict the probability of power outages in selected distribution circuits in Puerto Princesa City.

4.3 Ethical Considerations

The researchers will follow ethical standards and data privacy regulations throughout the process of the research. The data used in this study will include meteorological records obtained from Open-Meteo and vegetation data, as well as power service advisories published on the official website of the Palawan Electric Cooperative (PALECO). These advisories will be used to identify reported power interruptions and their affected areas.

In addition, the researchers will also apply for Ethical Clearance or Ethical Exemption from the University Research Office, if deemed necessary, to ensure compliance with institutional research ethics guidelines. The information gathered for analysis will be limited to details relevant to power interruptions, such as reported dates, times, affected locations, and event descriptions. Any incidental personal information appearing in the public advisories will be excluded from the model inputs and published results. The information will be used for the academic purposes outlined in this study. Moreover, the predictive model is intended to assist local utility providers and consumers in understanding outage likelihood and will not be used to make decisions that may cause harm or discriminatory treatment to any person or community.

4.4 Dataset Description

This study combines power-service advisories published on PALECO's official website with daily weather records and monthly vegetation data for Puerto Princesa City

from September 2022 to October 2026. The data are matched by date and affected area to examine patterns in reported power interruptions.

The advisories provide information about interruption dates, affected areas, and, when available, start times, restoration times, and causes. They are used to identify local interruptions and city-wide blackouts. A positive label indicates a reported interruption for an area and date; a negative label indicates no matching interruption in the available advisories. Records with unclear dates or affected areas are marked unknown. Previous published interruptions are also used to describe an area's service history.

Variable	Description
day_of_week	Day of the week extracted from the date and time record.
month	Month extracted from the date and time record.
season	Seasonal category based on the date, such as the dry or rainy season.

Table 4.1 List of Temporal features.

Temporal features will be extracted from the date and time records to capture time-based patterns in outage occurrence. Time-based variables such as day of week have been used in electric load forecasting studies, as demand related temporal indicators (Fu et al., 2022). In a study by Sivhugwana and Rangana (2025), it was found out that power outage levels can vary depending on the seasons. Therefore, this study will include day of week, month and season as engineered temporal features.

The daily weather data will be obtained from the Open-Meteo Historical Weather API for the period from September 2022 to October 2026. The variables include temperature, rainfall, wind speed, wind gusts, humidity, and weather code.

Variable	Description
temperature (°C)	Air temperature in Celsius
wind_speed	Average wind speed
rainfall	Amount of rain precipitation
precipitation	Total amount of all water falling from the sky
humidity	Air moisture level
weather_code	Encoded weather condition

Table 4.2 Weather features

- Temperature to see how heat spikes electricity use. When the city gets hotter, more people use cooling, which puts more stress on the local power supply.
- Wind Speed to find where branches or lines might get knocked down. High winds are a common cause of "tripping" in areas with a lot of trees.
- Rainfall to see how much rain it takes to cause a fault. Heavy rain can damage equipment or cause short circuits on exposed lines.
- Precipitation to measure the total water falling within the hour. Helps identify wet conditions that may be related to weather outages.
- Weather Code to identify the general weather conditions such as clear, cloudy or rainy.
- Relative Humidity to track the general weather systems moving through. These often give a heads-up that a storm is coming before the rain actually starts.

Variable	Description
rain_lag_1h	Rainfall value from one hour before the current record
rain_lag_3h	Rainfall value from three hours before the current record

rain_sum_last_3h	Total rainfall accumulated during the previous three hours
wind_gust_lag_1h	Wind gust value from one hour before the current record
max_wind_gust_last_6h	Highest wind gust recorded during the previous six hours
outage_count_last_24h	Number of outages recorded during the previous 24 hours.
rolling_outage_frequency_7d	Outage frequency over the past 7 days.

Table 4.3 Lagged weather and outage-related features.

The study includes “lagged” weather data, like a cumulative rainfall sum for the past three hours, to see if it has a delayed effect on the grid. It will also include lagged historical outage features, such as previous outage occurrence and recent outage frequency, to capture the recent outage behavior in the area. All of this is matched up hour-by-hour to create the final dataset. Each hour is then labeled as an “outage” or “normal” event, which the model uses to learn the relationship between weather and power reliability.

The monthly vegetation data will be processed using Google Earth Engine and summarized for each barangay. The available measures include NDVI and vegetation values within 5m, 15m, and 30m power-line buffers.

Variable	Description
ndvi_mean	Mean NDVI within the circuit buffer for the month, used as a baseline vegetation greenness indicator
veg_buffer_5m / veg_buffer_15m / veg_buffer_30m	Vegetation measured at different distances from the line
ndvi_trend_3mo	How fast vegetation has been growing over the last 3 months
ndvi_anomaly	How much greener or denser vegetation is than usual for that month

wind_veg_interaction	Wind speed combined with vegetation density
ndvi_landcover_masked	Vegetation greenness, counting only trees (not grass or crops)
ndvi_p90	Greenness of the densest vegetation spots near the circuit

Table 4.4 Vegetation-Related Variable

These variables describe the vegetation surrounding each barangays in Puerto Princesa City and how it changes over time. Additional variables track how quickly vegetation is growing and whether it is denser than usual for a given period. One variable isolates tree cover from grass and cropland, and another combines wind speed with vegetation density, since strong winds increase the risk posed by nearby vegetation. Together, these variables allow the model to capture when and where vegetation near the circuit may contribute to a power outage, particularly during the rainy season when plant growth and wind activity both increase.

Variable	**Description**
infra_type_encoded	One-hot encoded representation of the infrastructure type involved in each outage record
primary_poles_failure_rate_7d	Count of outages involving primary poles in the preceding 7 days
secondary_poles_failure_rate_7d	Count of outages involving secondary poles in the preceding 7 days
primary_lines_failure_rate_7d	Count of outages involving primary lines in the preceding 7 days
secondary_lines_failure_ratet_7d	Count of outages involving secondary lines in the preceding 7 days

transformer_failure_rate_30d	Count of transformer-involved outages in the past 30 days
infra_stress_index_7d	Total number of all outages across all five asset types in the past 7 days
mean_time_between_failures	The average time the distribution system operates between interruptions or outages.

Table 4.5 Equipment Reliability Proxy Features

The dataset utilized in this study also comprised of outage records across the electrical distribution circuit, grouped based on major infrastructure categories such as primary poles, secondary poles, primary lines, secondary lines, and transformers. This dataset catalogs how different assets and equipments have historically behaved under varying operational conditions.

Using this, the researchers plan to derive measures of structural reliability from the historical pattern of failures between infrastructure categories. The primary purpose of this data is to establish an empirical baseline of equipment and infrastructural vulnerability for the predictive model. This approach is grounded on the framework that grid reliability issues occur on the distribution side, where physical condition monitoring is rare. In addition, established distribution systems reliability studies have also demonstrated that component-specific fault rates and outage logs can be used to predict grid reliability indices (Normanyo & Diamenu, 2022).

4.5 Research Instruments

The primary data collection instruments to be used in this study are the PALECO data request, DOST-PAGASA, and Google Earth Engine. These instruments will be used to gather the outage, weather, and vegetation data for the proposed forecasting model.

PALECO Data Request

The PALECO data request will be used to gather historical outage and interruption records for Puerto Princesa City from September 2022 to 2026, including outage date and time, restoration time, affected barangays, circuit or recloser information where available, and narrative details of the interruption.

DOST-PAGASA Data Request

The DOST-PAGASA Data Request will be used to gather daily weather data for the study area. The weather variables include temperature, rainfall, precipitation, wind speed, wind gusts, humidity, and weather code.

Open-Mateo

This includes the same historical weather data (temperature, rainfall, precipitation, wind speed and gusts, humidity, weather code) obtained through the Open-Meteo Historical Weather API to supplement DOST-PAGASA records beyond **[period].**

Google Earth Engine

Google Earth Engine will be used to gather vegetation-related data through satellite imagery and NDVI. This will help estimate vegetation density near each barangays in Puerto Princesa City.

4.6 Model Development Instruments

In conducting this study, the researchers will follow a specific setup for pulling, cleaning, and analyzing the collected data.

Software. The researchers write and debug all Python scripts using Visual Studio Code (Microsoft, 2024). To explore the data and test small blocks of code, the researchers use Jupyter Notebook (Project Jupyter, 2024). For model training, the researchers utilize Google Colab to take advantage of its cloud computing capabilities, which provide faster processing for complex XGBoost computations (Google Research, 2024). Meanwhile, , the weather records were acquired through an official data request from DOST-PAGASA , which serves as the source of meteorological data based on the exact geographic coordinates of the study area (Philippine Atmospheric, Geophysical and Astronomical Services Administration [PAGASA], 2026).

Programming Language. The entire project is written in Python (Python Software Foundation, 2023). Python has emerged as the standard language for data science due to its versatile library ecosystem and its ability to handle large-scale time-series data

Libraries. The researchers rely on several specific tools to process the data:

- **Pandas** and **NumPy** are used for numerical processing, cleaning the raw files, filling in missing data, and making sure the timestamps on the power logs match up with the weather records (Harris et al., 2020; Mckinney, 2010).

- **Scikit-learn** is used for the initial setup. It helps the researchers scale the data and split it into training and testing groups. It also provides the metrics, like F1-score, to see how well the model is actually working. (Pedregosa et al., 2010).

- **XGBoost** is the main framework for the predictive model. It is very fast and particularly good at finding connections in this kind of data that simpler models might miss, as it utilizes a scalable tree-boosting system capable of modeling complex, non-linear relationships (Chen & Guestrin, 2016).

- **SHAP** will be used to interpret the trained XGBoost model by identifying the contribution of each input feature to the predicted outage likelihood, providing a game-theoretic framework for feature attribution (Lundberg & Lee, 20217).

- **Matplotlib** (Matplotlib 2024) and **Seaborn** (Waskom, 2021) are used to turn the final result into charts and graphs. This makes it easier to see how each weather factor, like wind or rain, actually affects the grid.

4.7 Research Procedures

Step 1. Data Collection

The researchers will collect historical outage and circuit interruption records from PALECO through a formal request. The requested data will cover the entirety of Puerto Princesa City from September 2022 to October 2026. The outage data will include outage date, outage time, outage duration, affected area, and available circuit-related information. The historical outage records will be categorized into Primary Pole per Distribution, Secondary Pole per Distribution, Primary Lines per Distribution System, Secondary Lines per Distribution System, and Transformer per Distribution System. Vegetation-related variables will be collected from Google Earth Engine (GEE) using satellite imagery, such as Sentinel-2. The researchers will use NDVI (Normalized Difference Vegetation Index) to identify and estimate vegetation coverage within each barangay area. The extracted vegetation data will then be summarized and matched with the outage, weather, and temporal dataset. Weather data will be collected from OpenMateo and the DOST-PAGASA historical weather data. The weather variables will include temperature, rainfall or precipitation, wind speed, wind gusts, humidity.

Step 2. Data Preparation

After collection, the datasets will be loaded into Python using Pandas. NumPy will be used for numerical processing (Harris et al, 2020). The process will check the datasets for missing values, duplicate records, inconsistent formats, and incorrect data types. The date and time fields will be converted into a standard datetime format. The outage records and weather records will be merged based on their date and time values.

Step 3. Feature Engineering

After the datasets are merged, feature variables extracted from historical outage data, weather data and vegetation-related data will be created using Pandas and Numpy. Temporal features will be extracted from the date and time column such as hour, day, month, and season. Lagged weather features will be created including previous rainfall, previous wind speed, previous wind gusts, or accumulated rainfall within a selected time window. Outage features will be extracted from past outage records such as previous outage occurrence and computed outage frequency. Vegetation-related features, such as vegetation density, will be derived from Google Earth Engine API to represent possible tree-related exposure in the study area. The target variable will be created by labeling each time period as either outage or no outage. **1** will represent an outage, while **0** will represent no recorded outage.

Step 4. Model Development

The prepared dataset will be divided chronologically into training, validation, and testing sets using Scikit-learn and Pandas. 70% of the datasets will be used for training using the XGBoost library, the next 15% will be used for validation and parameter tuning, and the 15% will be used for testing.

Step 5. Model Assessment

The final model will be evaluated using Scikit-learn metrics. Evaluation metrics such as precision, recall, F1-score, ROC-AUC will be used to measure the classification

performance of the model and Brier Score will assess the reliability of the predicted outage probabilities

SHAP will be used to interpret the trained XGBoost model to identify which feature is the most influential to the predicted output. This would help determine the importance of weather, temporal, outage history features in power outages and vegetation-related features in power outages.

Lastly, Matplotlib and Seaborn will be used to visualize the results. This includes the ROC curve, reliability diagram, and SHAP feature importance plots, to support the interpretation and reporting of model findings.

4.8 Data Analysis Plan

This study will utilize the following statistical methods and performance metrics to evaluate the result and performance of the developed model.

Metric	Definition	Use in the Study
Precision	Proportion of predicted outage events that are actual outage events.	Measures the rate of false alarms or events the model incorrectly flags as outage events.
Recall	Proportion of actual outage events correctly identified by the model.	Measures the rate of missed outages, the more operationally critical error type in utility grid management.
F1-Score	Harmonic mean of precision and recall, providing a single metric that balances both.	Primary classification performance metric; balances false positives and false negatives under potential class imbalance.
AUC-ROC	A threshold independent metric that quantifies a model's ability to discriminate between positive and negative classes	Measures the model's overall ability to discriminate between outage and non-outage events, independent of any specific probability threshold.

Brier Score	Mean squared error between predicted outage probabilities and observed binary outcomes.	Evaluates the accuracy of the model's probability estimates, and serves as the primary metric for both predictive and calibration quality.
Reliability Diagram	Plot of mean predicted probabilities against observed outage frequencies across binned prediction intervals.	Visually assesses probability calibration; a well-calibrated model produces points approximating the diagonal.
SHAP Values	Average marginal contribution of each feature to model predictions.	Identifies the most influential meteorological, temporal outage-history, and vegetation-related predictors of outage occurrence.

Table 4.5 Evaluation Metrics

Table 4.5 shows the statistical methods and performance metrics that will be used in the study. Given the limitations of raw accuracy under class imbalance, the performance of the developed model is assessed using the following theoretically grounded metrics:

Precision is the proportion of predicted outage events that are actual outage events:

$$precision = \frac{TP}{TP + FP}$$

Recall (sensitivity) is the proportion of actual outage events that are correctly identified:

$$recall = \frac{TP}{TP + FN}$$

For this study, recall is treated as the priority classification metric given the operational cost asymmetry identified in Section 3.3.

F1-Score is the harmonic mean of precision and recall, providing a single metric that balances both:

$$F_2 = \frac{Precision\ x\ Recall}{Precision + Recall}$$

The F1-score is particularly appropriate for imbalanced classification tasks because it is sensitive to performance on the minority class and is not inflated by a high prevalence of true negatives. For this study, a value of 0.60 or above under class-imbalanced conditions is treated as indicative of sufficient classification quality, reflecting the recall-prioritized evaluation standard applied in analogous early warning system research (Putra et al., 2025)

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) - measures the model's ability to discriminate between outage and non-outage events across all classification thresholds. An AUC-ROC of 1.0 indicates perfect discrimination; a value of 0.5 indicates performance equivalent to random chance. For this, an AUC-ROC of 0.75 or above is used as the minimum benchmark for acceptable discriminative performance, consistent with thresholds applied in weather-related outage forecasting studies (Sharma et al., 2023; Lee et al., 2025).

Reliability Diagram (also called a calibration curve) is used to assess the model's calibration empirically, in which predictions are grouped into equal-width bins and the mean predicted probability of each bin is plotted against the observed fraction of positive outcomes within that bin. A perfectly calibrated model produces points that lie along the diagonal. Systematic deviation above the diagonal indicates underconfidence; deviation below indicates overconfidence.

Brier Score measures the mean squared error between predicted probabilities and observed binary outcomes, with lower values indicating better probabilistic accuracy.

$$BS = \frac{1}{n} = \sum_{t=1}^{n} (p\hat{}t - yt)2$$

For this study, a Brier Score below 0.25 is considered indicative of reasonable probability estimate quality, consistent with the convention that a perfectly uninformative model produces a Brier Score approaching the base rate of the positive class.

4.9 Work Plan

To serve as a guideline for conducting the study, the researchers formulated a work plan that outlines the timeline of key activities and deliverables throughout the research period, from January to November 2026.

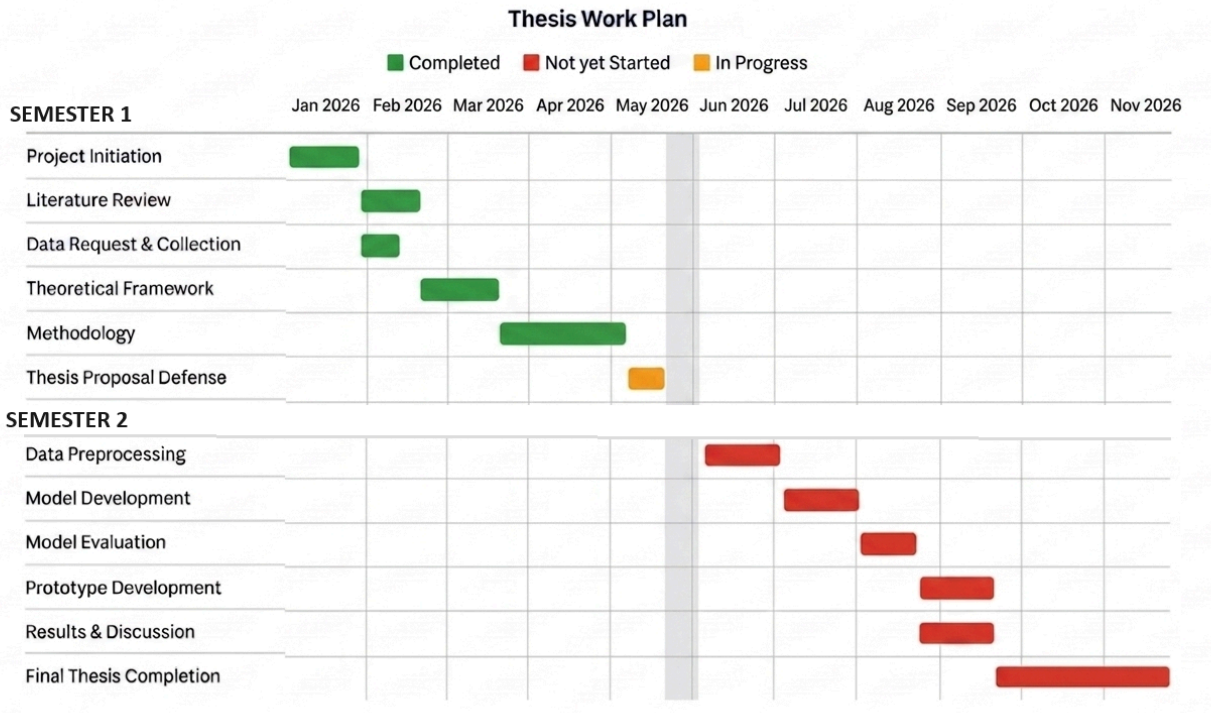


Figure 4.2 Gantt Chart

Figure 4.2 shows the work plan of the study, presented as a Gantt chart and divided into two semesters. The first semester covers the foundational phases of the research. The

study starts with Project Initiation in the final three weeks of January, where the research conceptualization, topic, objectives, scope and limitations are finalized in consultation with the thesis adviser. Literature Review and Data Request and Collection are parallel activities in early February, so they happen at the same time as the studies are being sourced and formal communication is initiated with the local distribution utility. The Theoretical Framework phase takes place from late February to late March, and the Methodology will span up to early May. The semester ends with the Thesis Proposal Defense followed by a short academic break before the second semester begins.

Activities for the second semester begins on early June and covers the technical and writing phases of the study. Data Preprocessing and Feature Engineering is carried out through early July, followed by Model Development from mid-July to early August. Model Evaluation succeeds this phase and runs through late August. Prototype Development and the writing of Results and Discussion are conducted in parallel from late August to late September, reflecting the interdependent nature of building the prototype and interpreting the model's outputs. The study concludes with the Final Thesis Completion phase, spanning from late September to November, which encompasses manuscript compilation, the final defense, incorporation of revisions, and submission of the completed thesis.

CHAPTER V

RESULT AND DISCUSSION

This chapter presents the results of developing and evaluating the XGBoost-based outage probability forecasting model for Puerto Princesa City. The results are organized according to the research objectives: dataset characteristics, model performance, probability calibration, SHAP-based feature interpretation, and the contribution of engineered features. The chapter ends with a discussion of the findings in relation to the reviewed literature.

5.1 Dataset Summary

After data preparation and merging, the final dataset covered September 2022 to September 2026 date] and contained [N – total records] records across [N – number of barangays/areas]. Of these, [N – outage] were labeled as outage (1) and [N – non-outage] as non-outage (0), giving an outage prevalence of [X.XX% – positive class rate].

Characteristic	Value
Observation period	[September 2022 – September 2026]
Total records	[N – total rows]
Outage records (y = 1)	[N – count] ([X.XX%])
Non-outage records (y = 0)	[N – count] ([X.XX%])
Class ratio (non-outage : outage)	[X : 1 – ratio]
scale_pos_weight applied	[X.XX – count(y=0)/count(y=1)]
Records removed during cleaning	[N – duplicates/missing]
Total features after engineering	[N – feature count]

Table 5.1 Summary of the Final Dataset

Set	Proportion	Period	Records	Outage prevalence
Training	70%	[date start – end]	[N]	[X.XX%]
Validation	15%	[date start – end]	[N]	[X.XX%]
Testing	15%	[date start – end]	[N]	[X.XX%]

Table 5.2 Chronological Data Split

The data showed a [severe / moderate / mild] class imbalance, with positive outages making up only [X.XX%] of all records. This confirms the concern in Section 3.3 that accuracy is not an appropriate primary metric, and the evaluation therefore relies on recall, precision, F1-score, AUC-ROC, and Brier Score.

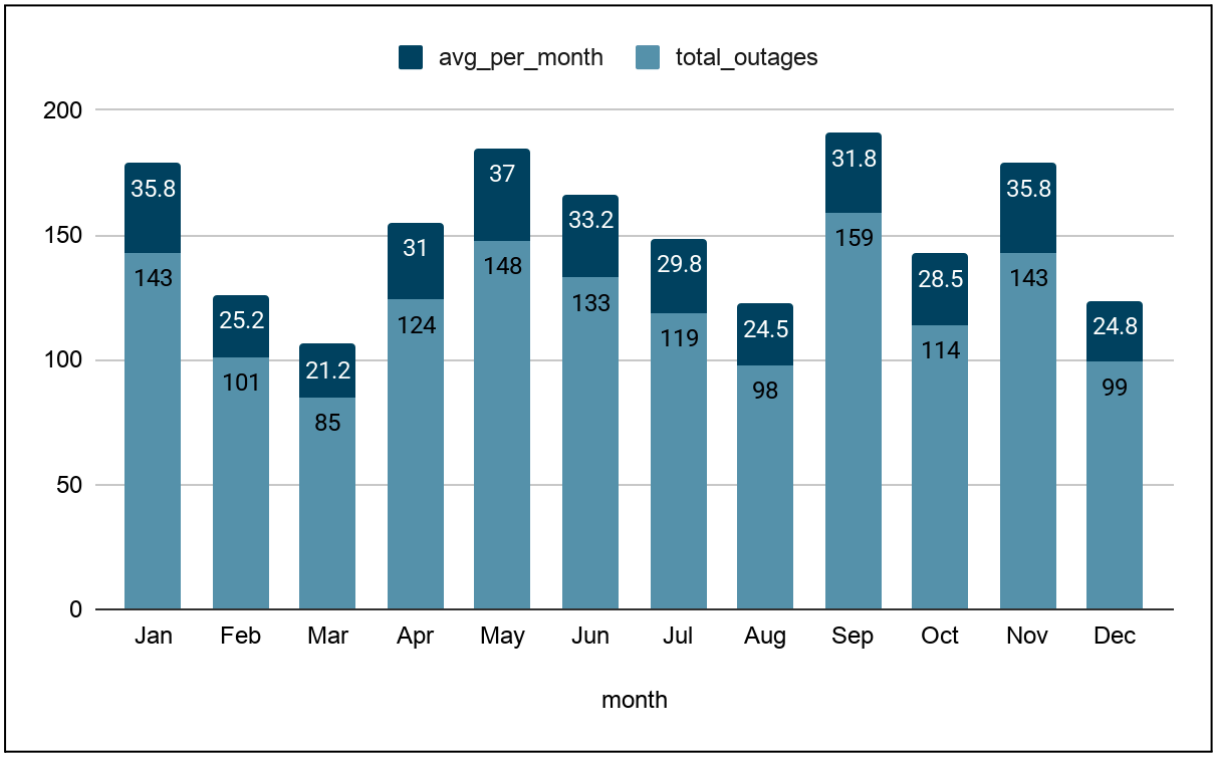


Figure 5.1: Monthly total outages and average counts

[Describe any notable pattern here, e.g., outages peaked during [month/season - period], and [infrastructure type – e.g., secondary lines] accounted for [X%] of recorded outages.]

5.2 Feature Engineering Results

[N – feature count] features were engineered and used for model training, grouped as shown in Table 5.3.

Feature group	Number of features	Examples
Temporal	[N]	day_of_week, month, season [+ others retained]
Meteorological	[N]	temperature, wind_speed, rainfall
Lagged weather	[N]	rain_lag_1h, rain_sum_last_3h, max_wind_gust_last_6h
Outage history	[N]	outage_count_last_24h, rolling_outage_frequency_7d
Infrastructure reliability	[N]	primary_poles_failure_rate_7d, infra_stress_index_7d
Vegetation	[N]	ndvi_mean, ndvi_anomaly, wind_veg_interaction
Total	**[N]**	

Table 5.3 Feature Groups Used in the Model

[State any features dropped (e.g., due to high correlation, missingness, or zero variance), the reason, and how missing values were handled.]

5.3 Model Development and Hyperparameter Tuning

Hyperparameter	Search range	Final value
n_estimators	[range]	[value]
max_depth	[range]	[value]
learning_rate	[range]	[value]
subsample	[range]	[value]
colsample_bytree	[range]	[value]

min_child_weight	[range]	[value]
gamma	[range]	[value]
reg_lambda / reg_alpha	[range]	[value]
scale_pos_weight	–	[value]

Table 5.4 Final Hyperparameters

The XGBoost model was trained on the training set and tuned using the validation set. [Describe the tuning method – e.g., grid search / random search / Bayesian optimization] was used, with [metric – e.g., F1-score / AUC-PR] as the tuning objective. Early stopping was applied after [N – rounds] rounds without improvement.

5.4 Predictive Performance of the Model (Objective 2)

Metric	Validation	Test	Benchmark (Section 4.8)	Met
Precision	[0.XX]	[0.XX]	–	–
Recall	[0.XX]	[0.XX]	Priority metric	–
F1-score	[0.XX]	[0.XX]	≥ 0.60	[Yes / No]
AUC-ROC	[0.XX]	[0.XX]	≥ 0.75	[Yes / No]
Brier Score	[0.XXX]	[0.XXX]	< 0.25	[Yes / No]

Table 5.5 Model Performance on Validation and Test Sets

On the test set, the model achieved an AUC-ROC of [0.XX], {exceeding / falling below} the 0.75 benchmark. This indicates {strong / acceptable / limited} ability to discriminate between outage and non-outage hours. The F1-score of [0.XX] {met / did not meet} the 0.60 threshold, with a precision of [0.XX] and a recall of [0.XX]. This means that of all actual outage hours in the test set, the model correctly identified [XX%], while [XX%] of the hours it flagged as outages were genuine.

Figure 5.2 ROC Curve of the XGBoost Model *[insert figure – ROC curve with AUC annotation]*

	Predicted: No Outage	**Predicted: Outage**
Actual: No Outage	[TN]:	[FP]:
Actual: Outage	[FN]:	[TP]:

Table 5.6 Confusion Matrix on the Test Set (threshold = [0.XX – decision threshold])

The model produced [FN] false negatives (missed outages) and [FP] false positives (false alarms). Because missed outages are operationally more costly (Section 3.3), the decision threshold was set to [0.XX – threshold] to {favor recall / balance precision and recall}. [Explain how the threshold was chosen, e.g., maximizing F1 on the validation set or a minimum recall target.]

5.5 Probability Calibration (Objective 2)

Figure 5.3 Reliability Diagram of Predicted Outage Probabilities *[insert figure – calibration curve with diagonal reference]*

The Brier Score on the test set was [0.XXX]. For reference, a model that always predicts the base rate of [X.XX%] would obtain a Brier Score of approximately [0.XXX – p(1−p)], so the model {improves on / does not improve on} this baseline by [XX%]. The reliability diagram shows that predicted probabilities {closely follow the diagonal / fall above the diagonal (underconfident) / fall below the diagonal (overconfident)}, particularly in the [low / middle / high] probability range ([0.XX–0.XX]). [If calibration was applied, e.g., isotonic regression or Platt scaling, report the Brier Score before ([0.XXX]) and after ([0.XXX]).]

This indicates that the outage likelihood percentages produced by the model {can / can only partially} be interpreted as risk levels, as described in Section 3.2.

5.6 SHAP-Based Feature Importance (Objective 3)

5.6.1 Global Feature Importance

Figure 5.4 SHAP Summary Bar Plot (Mean |SHAP value|) *[insert figure]*

Rank	Feature	Group	Mean |SHAP|	Direction of effect
1	[feature]	[group]	[0.XXX]	[↑ / ↓ outage risk with higher value]
2	[feature]	[group]	[0.XXX]	[ ]
3	[feature]	[group]	[0.XXX]	[ ]
…	…	…	…	…
10	[feature]	[group]	[0.XXX]	[ ]

Table 5.8 Top [N/10] Features by Mean Absolute SHAP Value

The most influential feature was [feature – top feature], followed by [feature] and [feature]. [Interpret: e.g., recent outage activity (outage_count_last_24h) was the strongest predictor, suggesting that outages cluster in time.]

Figure 5.5 SHAP Beeswarm Plot *[insert figure]*

[Interpret how feature values relate to SHAP values, e.g., high values of rain_sum_last_3h push predictions toward higher outage probability.]

5.6.2 Contribution by Feature Group

Feature group	Sum of mean |SHAP|	Share of total (%)

Outage history	[0.XXX]	[XX%]
Infrastructure reliability	[0.XXX]	[XX%]
Meteorological (incl. lagged)	[0.XXX]	[XX%]
Temporal	[0.XXX]	[XX%]
Vegetation	[0.XXX]	[XX%]

Table 5.9 Aggregated SHAP Contribution by Feature Group

[Discuss which groups dominated, and what this implies for the island-based Puerto Princesa grid, where outages occur under both routine and weather-driven conditions.]

5.6.3 Feature-Level Interpretation

Figure 5.6 SHAP Dependence Plots for Selected Features *[insert figure – e.g., rainfall, wind speed, ndvi_anomaly]*

- **Meteorological:** [e.g., Outage probability increased sharply when rainfall exceeded [X mm] / wind gusts exceeded [X m/s].]
- **Temporal:** [e.g., Outage risk was higher during {rainy / dry} season months ([months]) and on {day of week}.]
- **Vegetation:** [e.g., wind_veg_interaction contributed {positively / negligibly}, suggesting that {vegetation-related risk is amplified by wind / vegetation had limited influence at the available resolution}.]
- **Infrastructure:** [e.g., Recent failures in {asset type – e.g., secondary lines} were associated with higher outage likelihood.]

5.7 Contribution of Lagged and Cumulative Outage Features (Objective 4)

To assess the impact of the engineered features, the model was retrained with different feature sets under identical hyperparameters and data splits.

Feature set	Precision	Recall	F1	AUC-ROC	Brier
A. Base features only (raw weather + temporal)	[ ]	[ ]	[ ]	[ ]	[ ]
B. A + lagged weather features	[ ]	[ ]	[ ]	[ ]	[ ]
C. A + outage history features (lag/rolling)	[ ]	[ ]	[ ]	[ ]	[ ]
D. A + vegetation features	[ ]	[ ]	[ ]	[ ]	[ ]
E. A + infrastructure reliability features	[ ]	[ ]	[ ]	[ ]	[ ]
F. All features (final model)	[ ]	[ ]	[ ]	[ ]	[ ]

Table 5.10 Ablation Results on the Test Set

Adding lagged weather features changed AUC-ROC by [+/− 0.XX] and F1-score by [+/− 0.XX]. Adding outage history features (outage_count_last_24h, rolling_outage_frequency_7d) changed AUC-ROC by [+/− 0.XX] and F1-score by [+/− 0.XX]. [Interpret which engineered group contributed the most, and whether the improvement is meaningful or marginal.]

5.8 Discussion

- **Relation to the reviewed literature.** The model's AUC-ROC of [0.XX] is {comparable to / lower than / higher than} the results reported in related studies, such as [Saidi et al. (2021) – state their metric], [Putra et al. (2025) – state their metric], and [Lee et al. (2025) – state their metric]. [Note that direct comparison is limited by differences in data, region, and class balance.]
- **Weather and historical data.** Consistent with Wang et al. (2022) and Putra et al. (2025), combining meteorological and outage-history features {improved / did not clearly improve} performance over using either alone. This {supports / only partially supports} the argument in Section 2.3 that practical, commonly available data can be sufficient for outage forecasting. [Relate to Talukdar & Toweh (2024) if weather alone was insufficient.]
- **Role of vegetation and infrastructure features.** [Discuss whether the vegetation (NDVI-based) and infrastructure reliability proxy features contributed meaningfully, and why. Note any limitation of monthly NDVI resolution against hourly outage labels.]
- **Routine versus weather-driven outages.** [Discuss whether the model performed differently under normal conditions versus rain/wind-heavy periods, if analyzed, e.g., recall of [0.XX] during rainy months versus [0.XX] during dry months.]
- **Practical implications for PALECO.** [e.g., The model can rank hours or areas by outage risk, supporting crew pre-positioning during periods where predicted probability exceeds [0.XX]. The reliability diagram suggests the probabilities {can / cannot yet} be used directly as risk levels.]
- **Observed limitations.** [e.g., limited number of outage events, inconsistent log entries, coarse weather resolution (daily vs. hourly), no actual line/asset age data, rare typhoon events underrepresented.]

CHAPTER VI

CONCLUSION AND RECOMMENDATION

6.1 Conclusions

Based on the results of the study, the following conclusions were drawn:

1. **Model development.** Historical outage logs, meteorological variables, temporal features, vegetation-related variables, and infrastructure reliability proxy features were successfully integrated to develop an XGBoost-based outage probability forecasting model for Puerto Princesa City. The final dataset contained [N] records and [N] engineered features.
2. **Predictive performance.** On the held-out test set, the model achieved an AUC-ROC of [0.XX], F1-score of [0.XX], precision of [0.XX], recall of [0.XX], and Brier Score of [0.XXX]. The model therefore {met / partially met / did not meet} the benchmarks of AUC-ROC ≥ 0.75, F1-score ≥ 0.60, and Brier Score < 0.25 set in this study. The reliability analysis showed that the predicted probabilities were {well calibrated / overconfident / underconfident}.
3. **Feature importance.** SHAP analysis showed that [top 3 features – feature names] were the most influential predictors of outage likelihood. By feature group, [group – e.g., outage history] contributed the most ([XX%]), followed by [group] ([XX%]) and [group] ([XX%]). Vegetation-related variables {contributed meaningfully / contributed minimally}, with [feature – e.g., ndvi_anomaly] being the most relevant among them.
4. **Engineered features.** Lagged weather and cumulative outage frequency features {improved / did not substantially improve} model performance, raising AUC-ROC

from [0.XX] to [0.XX] and F1-score from [0.XX] to [0.XX] relative to the base feature set.

5. **Overall.** The study {demonstrates / suggests} that a tree-based ensemble model using practical, locally available data can {effectively / reasonably} forecast outage probability in an isolated island grid such as Puerto Princesa City, covering both routine and weather-influenced outages.

6.2 Recommendations

For the utility (PALECO) and local stakeholders

1. [e.g., Use the predicted outage probabilities as a risk-ranking tool for crew scheduling and resource pre-positioning, starting with periods or areas above [0.XX] probability.]
2. [e.g., Standardize outage logging (consistent timestamps, restoration times, asset type, and cause fields) to improve data quality for future model updates.]
3. [e.g., Prioritize vegetation management and inspection in areas and seasons identified by SHAP as high-risk.]
4. [e.g., Record asset-level information such as equipment age, maintenance history, and exact pole/line IDs to enable more precise infrastructure features.]

For model and data improvement

5. Retrain and revalidate the model periodically as new outage and weather data become available, and monitor calibration drift.
6. [e.g., Use higher-resolution weather data (hourly station data, rainfall radar, or satellite products) and higher-frequency vegetation indices.]
7. [e.g., Incorporate additional data such as lightning strikes, load/demand data, or generation shortfall records, which are major drivers in island systems.]

For future researchers

8. Compare XGBoost with other algorithms (e.g., LightGBM, CatBoost) and sequence-based models (e.g., LSTM) on the same dataset.
9. Extend the model to additional distribution circuits or other island grids to test generalizability.
10. Explore outage duration prediction or multi-hour-ahead forecast horizons, and handle imbalance using alternative techniques (e.g., SMOTE, focal loss) for comparison with scale_pos_weight.
11. **Prototype/system development.** Since this study was model-centric, future work may develop a deployable prototype, such as a dashboard or advisory tool that ingests live weather forecasts and displays outage probability by barangay for utility operators, with the trained model as its prediction engine.

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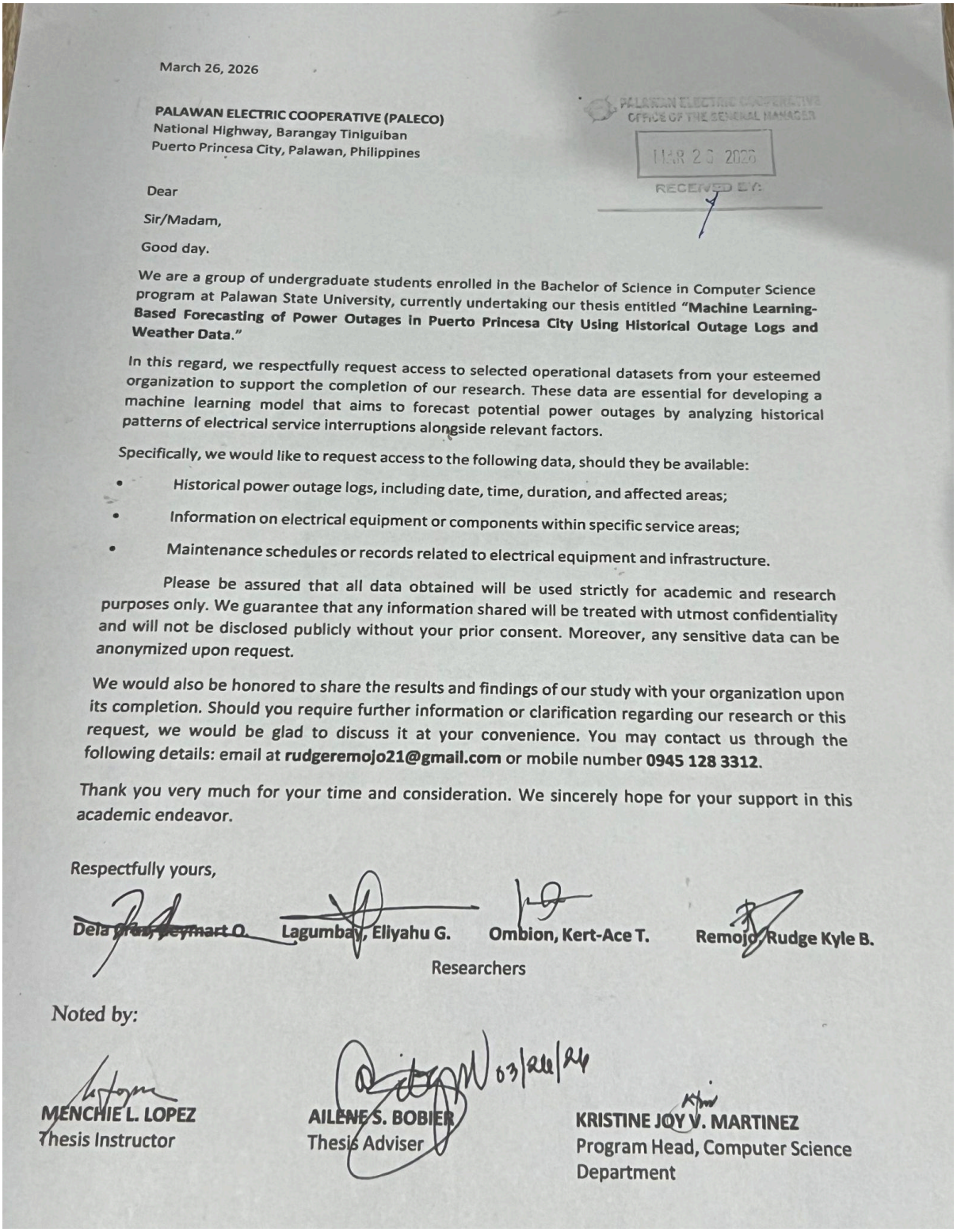
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APPENDICES

Appendix A: Request Letter



March 26, 2026

PALAWAN ELECTRIC COOPERATIVE (PALECO)
National Highway, Barangay Tinlguiban
Puerto Princesa City, Palawan, Philippines

PALAWAN ELECTRIC COOPERATIVE
OFFICE OF THE GENERAL MANAGER
MAR 2 6 2026
RECEIVED BY:

Dear

Sir/Madam,

Good day.

We are a group of undergraduate students enrolled in the Bachelor of Science in Computer Science program at Palawan State University, currently undertaking our thesis entitled **"Machine Learning-Based Forecasting of Power Outages in Puerto Princesa City Using Historical Outage Logs and Weather Data."**

In this regard, we respectfully request access to selected operational datasets from your esteemed organization to support the completion of our research. These data are essential for developing a machine learning model that aims to forecast potential power outages by analyzing historical patterns of electrical service interruptions alongside relevant factors.

Specifically, we would like to request access to the following data, should they be available:

- Historical power outage logs, including date, time, duration, and affected areas;
- Information on electrical equipment or components within specific service areas;
- Maintenance schedules or records related to electrical equipment and infrastructure.

Please be assured that all data obtained will be used strictly for academic and research purposes only. We guarantee that any information shared will be treated with utmost confidentiality and will not be disclosed publicly without your prior consent. Moreover, any sensitive data can be anonymized upon request.

We would also be honored to share the results and findings of our study with your organization upon its completion. Should you require further information or clarification regarding our research or this request, we would be glad to discuss it at your convenience. You may contact us through the following details: email at **rudgeremojo21@gmail.com** or mobile number **0945 128 3312**.

Thank you very much for your time and consideration. We sincerely hope for your support in this academic endeavor.

Respectfully yours,

Dela Cruz, Jeymart O. Lagumbay, Eliyahu G. Ombion, Kert-Ace T. Remojo, Rudge Kyle B.
Researchers

Noted by:

03/26/26

MENCHIE L. LOPEZ
Thesis Instructor

AILENE S. BOBIER
Thesis Adviser

KRISTINE JOY V. MARTINEZ
Program Head, Computer Science Department

A formal letter of request to Palawan Electric Cooperative (PALECO), detailing the intent for the collection of the specified data.

Appendix B: Data Request Price



DATA REQUEST

DATE: **May 25, 2026**

DATA REQUEST LIST FOR: **Note**: Proceed to Cashier for the payment of requested data.

R. Dela Cruz, E. Lagumbay, K. Ombion, R. Remojo

Payment of Requested Data/Information. Prices are based on PALECO's Board Resolution 124, S. 2018

NO.	REQUESTED DATA	CATEGORY	RATE	COVERAGE/ LOCATION	AMOUNT (PHP)
1	**GIS Mapping data of Poles, lines, and electrical assets**	Fault Current per specific location	10,000.00	Wescom Circuit	10,000.00
	Historical Outage and Interruption (2021-2025)	Primary Pole per Distribution	2,000.00		2,000.00
		Secondary Pole per Distribution	2,000.00		2,000.00
		Primary Lines per Distribution System	2,000.00		2,000.00
		Secondary Lines per Distribution System	2,000.00		2,000.00
		Transformer per Distribution System	2,000.00		2,000.00
		Loads per month per Distribution System	2000*12*5 = 120,000		120,000.00
		****Nothing Follows****			140,000.00
		12% VAT			16,800.00
		TOTAL			**156,800.00**
		Student Discount			50%
		GRAND TOTAL			**78,400.00**

Note: *Only data regarding the Historical Outage and Interruption (2021-2025), and Infrastructure categories (excluding the load per month per distribution system), will be purchased and used in this study due cost related constraints.*

Appendix C: Certification of Readiness for Proposal Defense

PSU-ACA-008

Republic of the Philippines
PALAWAN STATE UNIVERSITY
Puerto Princesa City

College of Sciences
Department of Computer Science

Certification of Readiness for Proposal Defense

RONALD EDILBERTO A. ONA
Dean
This College

Sir:

This is to certify that **Reymart Dela Cruz, Eliyahu Lagumbay, Kert Ace Ombion, & Rudge Kyle Remojo** are ready to defend their research study, **Power Outage Probability Forecasting with XGBoost and SHAP-Based Feature Analysis Using Historical and Meteorological Data**

It is requested that oral proposal defense be scheduled.

AILENE BOBIER
Research Adviser

Schedule for Oral Proposal Defense:

Date & Time: **May 18, 2026 10:00 - 12:00 PM**

Venue: **Co-Working Space, Student Innovation Park Building**

Recommending Approval:

MENCHIE LOPEZ
Research Instructor

Approved

RONALD EDILBERTO A. ONA
Dean

Copy Furnished:
Research Adviser
Research Coordinator
Department Chairperson
Dean
Proponents

Appendix D: Certification of Readiness for Proposal Defense

PSU-ACA-010

Republic of the Philippines
PALAWAN STATE UNIVERSITY
Puerto Princesa City

College of Sciences

May 18, 2026
Date

Request for Proposal Defense

RONALD EDILBERTO A. ONA. PhD
The Dean
This College

Sir / Madam:

The undersigned request your approval for the schedule of my/our proposal defense on **May 18, 2026** for my/our thesis/dissertation/special problem paper entitled: **Power Outage Probability Forecasting with XGBoost and SHAP-Based Feature Analysis Using Historical and Meteorological Data** and that the following Faculty are requested as Panel of Examiners.

Chairman	**KAYE JANIECKA PARCON**
Member	**DIVINE CAABAY**
Member	**VINCENT ADRIAN ABIAN**

In relation to this, I/we have provided _4_ copies of my/our thesis/dissertation/special problem paper for distribution to the Panel of Examiners.

Truly yours,

REYMART DELA CRUZ
ELIYAHU LAGUMBAY
KERT ACE OMBION
RUDGE KYLE REMOJO

Student's Name & Signature

Noted:

AILENE BOBIER
Research Adviser

Recommending Approval:

MENCHIE LOPEZ
Research Instructor/Professor/Critic Reader

Approved:

RONALD EDILBERTO A. ONA. PhD
Dean